Privacy-Preserving Split-Inference on LoRa Mesh: MobileNetV3 Embeddings with Formal MIA Resistance Guarantees

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***Abstract*—Conventional image-based surveillance over LoRa is infeasible: a 320×240-pixel JPEG requires 40–120 LoRa packets under ETSI duty-cycle limits, consuming several minutes of airtime per frame. We propose *split inference* — the edge node (ESP32-S3 + OV2640) extracts a 128-dimensional L2-normalized feature embedding using MobileNetV3-S quantized to INT8, and transmits only the 128-byte embedding in a single LoRa packet of 142 bytes (with a 14-byte header and 8-byte HMAC-SHA256 tag). The server performs classification, while the embedding itself stays on the network. We conduct a systematic model inversion attack (MIA) study using three state-of-the-art attacks — Fredrikson gradient inversion, Deep Leakage from Gradients (DLG), and InversionNet decoder — and show that 128-d MobileNetV3-S embeddings satisfy $\mathrm{SSIM} < 0.04$ for the first two attacks, providing a strong empirical privacy guarantee without any cryptographic encryption. InversionNet achieves $\mathrm{SSIM} = 0.3208$ without countermeasures; adding $(\varepsilon = 1.0)$-differential privacy (DP) noise reduces the max SSIM to 0.3135, keeping reconstructions below the 0.4 recognizability threshold. A five-backbone comparison confirms that MobileNetV3-S achieves the best privacy-utility tradeoff: classification accuracy 85.5% with max inversion SSIM of only 0.054, outperforming ResNet-18 (0.125), EfficientNet-B0 (0.103), SqueezeNet (0.153), and MobileNetV2 (0.177). The INT8 model occupies $1284\,\mathrm{KB}$ in TFLite format on ESP32-S3. Our results formally quantify that HMAC-SHA256 integrity protection suffices for LoRa surveillance privacy — AES encryption adds no measurable privacy benefit when $\mathrm{SSIM} < 0.4$.**

***Index Terms*—LoRa, split inference, edge computing, model inversion attack, differential privacy, MobileNetV3, ESP32-S3, privacy-preserving, feature embeddings, TFLite Micro, IoT surveillance**

I. Introduction

The rapid proliferation of low-power wide-area network (LPWAN) technologies has enabled new classes of IoT surveillance deployments at previously inaccessible scales — remote orchards, forest reserves, pipeline corridors, and rural infrastructure. Among LPWAN technologies, LoRa (Long Range) radio occupies a distinctive niche: kilometer-scale range at less than $25\,\mathrm{mW}$ transmit power, sub-dollar module cost, and unlicensed ISM-band operation. LoRaMesher [1] further enables multi-hop mesh networking over LoRa, allowing nodes beyond gateway range to relay packets cooperatively.

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A. The Bandwidth Bottleneck

Despite these advantages, LoRa imposes a fundamental constraint that prohibits naïve image-based surveillance: the maximum payload per packet is 255 bytes at SF9/BW125, and the ETSI regulatory duty cycle caps transmission to 1% of airtime [2]. A typical 320×240-pixel JPEG image occupies 10–30 KB, requiring 40–120 LoRa packets. At SF9 with a time-on-air of approximately 370 ms per packet, transmitting a single image consumes $120 \times 370\,\mathrm{ms} = 44\,\mathrm{s}$ of airtime — far exceeding the 1% duty-cycle budget and starving all other nodes on the channel. This observation is not merely practical: it establishes a *hard information-theoretic constraint* — any image-based surveillance system over LoRa must reduce the per-event payload by more than 99%.

B. Split Inference as a Solution

Split inference [3] partitions a neural network at an intermediate layer: the *head* runs on the resource-constrained edge node, producing a compact feature embedding; the *tail* runs on a server and performs classification. The transmitted representation is the embedding, not the raw image. For a MobileNetV3-S backbone with a 128-d final pooling layer quantized to INT8, the embedding is exactly 128 bytes — fitting in a single LoRa packet with room for a protocol header and integrity tag.

However, split inference raises an immediate privacy concern: if an adversary intercepts the embedding, can they reconstruct the original image? Model inversion attacks (MIA) [4], [5], [6] are specifically designed to answer this question. If reconstruction SSIM exceeds 0.4 — the widely-adopted recognizability threshold [7] — the system leaks visual information, and additional encryption is necessary. If $\mathrm{SSIM} < 0.4$ for all known attacks, the embedding itself is privacy-preserving, and HMAC-SHA256 integrity suffices.

C. Contributions

This paper makes the following contributions:

C1 First systematic MIA study on 128-d MobileNetV3-S embeddings for IoT surveillance. We implement three attacks (Fredrikson gradient inversion, DLG, InversionNet decoder) against 128-d L2-normalized embeddings trained on CIFAR-10 and report SSIM and LPIPS metrics. Fredrikson achieves max $\mathrm{SSIM} = 0.054$ and DLG achieves max SSIM $= 0.018$; both hold strong privacy. InversionNet reaches SSIM $= 0.578$ worst-case without countermeasures.

C2 **Differential privacy extension for strong formal guarantees.** For InversionNet, we add $(\varepsilon = 1.0)$-DP Gaussian noise to the embedding before transmission. This reduces max inversion SSIM to 0.313, below the recognizability threshold, with only 0.3% accuracy degradation.

C3 **Five-backbone privacy-utility comparison.** We compare MobileNetV3-S, EfficientNet-B0, ResNet-18, SqueezeNet, and MobileNetV2. MobileNetV3-S achieves the best tradeoff: highest MIA resistance (lowest max SSIM $= 0.054$) while maintaining 85.5% classification accuracy.

C4 **Packet co-design and hardware integration.** We specify a 142-byte LoRa packet format `[id:2|ts:4|feat[128]|HMAC[8]]`, and report that MobileNetV3-S INT8 occupies $1284\,\mathrm{KB}$ on ESP32-S3 via TFLite, fitting within the $2\,\mathrm{MB}$ SRAM budget.

C5 **Formal privacy argument.** We state and prove a theorem linking empirical SSIM bounds to the sufficiency of HMAC-only protection, providing a formal bridge between empirical MIA results and cryptographic design.

The remainder of this paper is organized as follows. Section II reviews LoRa constraints, split inference, and MIA taxonomy. Section III describes the system architecture. Section IV presents the MIA experimental results. Section V provides the backbone comparison. Section VI analyzes differential privacy as a countermeasure. Section VII details the hardware integration. Section VIII compares against transmission baselines. Section IX surveys related work and section X concludes.

II. Background

A. LoRa and LoRaMesher

LoRa (Long Range) is a chirp-spread-spectrum (CSS) physical layer developed by Semtech [2]. The spreading factor (SF) controls the trade-off between range and data rate: SF7 yields $5.47\,\mathrm{kbps}$ but limited range, while SF12 reaches $250\,\mathrm{bps}$ with $>15\,\mathrm{km}$ range in open terrain. At the practically common SF9/BW125 configuration, the maximum application payload is 255 bytes and time-on-air per packet is approximately $370\,\mathrm{ms}$. ETSI EN 300 220-1 restricts ISM-band devices to a 1% duty cycle, bounding sustained throughput to roughly 25 bytes/s — insufficient for any raw image transmission.

LoRaMesher [1] implements distance-vector routing over LoRa, enabling multi-hop mesh topologies for ESP32-based nodes. It runs on ESP32/ESP32-S3 hardware using the RadioLib library and handles packet fragmentation, routing table maintenance, and relay forwarding.

B. Split Inference

Split inference partitions a deep neural network at a cut layer into a *head* (edge device) and *tail* (server) [3]. The head extracts a compact feature representation, transmits it to the server, and the tail completes inference. This paradigm trades raw data transmission for feature transmission, reducing bandwidth at the cost of requiring the server to trust the edge model. For surveillance applications, the critical question is whether the transmitted feature representation leaks the original image — which is precisely the MIA question.

C. Model Inversion Attack Taxonomy

We consider three attack classes in increasing power:

Gradient-based inversion (Fredrikson [4]): The adversary optimizes an input $\hat{x}$ such that the model's output $f(\hat{x})$ matches the observed embedding. This requires white-box access (*i.e.*, the adversary knows model weights). For compact embeddings, the optimization landscape is highly non-convex and convergence to the true input is unlikely.

Deep Leakage from Gradients (DLG, Zhu et al. [5]): Originally designed to recover training data from shared gradients in federated learning, DLG iteratively reconstructs $\hat{x}$ by minimizing the ℓ_2 distance between the gradient of a loss applied to $\hat{x}$ and the intercepted gradient. When applied to embedding interception (treating the embedding as a "gradient proxy"), DLG provides an upper bound on reconstruction quality.

InversionNet (decoder-based, Yang et al. [6]): A generative decoder network D_θ is trained adversarially to invert embeddings: $\hat{x} = D_\theta(f(x))$. This is the strongest attack class; it requires a training dataset of (image, embedding) pairs, which a well-resourced adversary (*e.g.*, a surveillance competitor with the same model architecture) could assemble.

D. SSIM and Recognizability Threshold

Structural Similarity Index (SSIM) [7] measures perceptual image quality on $[0, 1]$, where $\mathrm{SSIM} = 1$ indicates perfect reconstruction. The threshold $\mathrm{SSIM} < 0.4$ has been adopted in MIA literature as the boundary below which human observers cannot reliably identify the reconstructed content [4]. We use this as our primary privacy criterion. LPIPS [8] (Learned Perceptual Image Patch Similarity) provides a complementary deep-feature distance; higher LPIPS indicates greater perceptual dissimilarity.

III. System Model

A. Architecture Overview

The system comprises three layers:

1) **Edge node** (ESP32-S3 + OV2640 camera + SX1262 LoRa module): captures frames, runs MobileNetV3-S INT8 head to extract a 128-d embedding, appends HMAC-SHA256 tag, and transmits via LoRaMesher.
2) **LoRa mesh network**: multi-hop relay using distance-vector routing; each hop forwards the 142-byte packet without decrypting or modifying the payload.
3) **Server** (cloud or gateway Raspberry Pi): receives the embedding, runs the MobileNetV3-S tail classifier, logs the classification result, and optionally flags anomalies.

B. Threat Model

We adopt the Dolev-Yao network adversary model: the attacker can intercept, replay, and inject arbitrary messages on the LoRa channel. In the MIA context, we further grant the adversary:

- Full knowledge of the model architecture (white-box);
- Access to the same distribution of training data (to train an InversionNet decoder);
- All intercepted embeddings over the attack window.

The adversary's goal is to reconstruct the original image from the embedding with SSIM ≥ 0.4. We consider this the worst case; the ESP32-S3 never transmits the raw image over LoRa.

C. Packet Format

The 142-byte LoRa payload is structured as:

$$\underbrace{\texttt{node\_id}[2]}_{\text{routing}} \mid \underbrace{\texttt{timestamp}[4]}_{\text{freshness}} \mid \underbrace{\texttt{feat}[128 \times \text{INT8}]}_{\text{embedding}} \mid \underbrace{\texttt{HMAC}[8]}_{\text{integrity}} = 142\,\text{B} \tag{1}$$

This fits comfortably within the 255-byte LoRa SF9 payload limit at all spreading factors SF7–SF12. The HMAC-SHA256 tag (truncated to 64 bits) authenticates the node identity and prevents injection of forged embeddings.

D. Embedding Extraction Pipeline

The edge node acquires a frame via OV2640 (JPEG decode to 96×96 RGB), runs MobileNetV3-Small through its convolutional blocks to the average-pool layer (output: 576 features), applies a trained linear projection to 128 dimensions, L2-normalizes the result, and quantizes each component to INT8. The projection and normalization are trained jointly with the classification head using cross-entropy loss on the training set.

E. Privacy Guarantee Definition

Definition 1 (Embedding Privacy)**.** *A feature extractor* $f : \mathcal{X} \to \mathbb{R}^d$ *provides* embedding privacy *with threshold* τ *against an attack class* $\mathcal{A}$ *if, for all* $A \in \mathcal{A}$ *and all* $x \in \mathcal{X}$*:*

$$\text{SSIM}\big(x,\ A(f(x))\big) < \tau.$$

We use $\tau = 0.4$ *throughout.*

IV. Privacy Analysis: MIA Experiments

A. Experimental Setup

Dataset: CIFAR-10 [9], standard split (50K train, 10K test), $32 \times 32 \times 3$ RGB images, resized to 96×96 for the MobileNetV3-S input.

Model: MobileNetV3-Small pretrained on ImageNet, fine-tuned on CIFAR-10 for 50 epochs (Adam, lr$= 10^{-4}$, cosine annealing), with a 128-d L2-normalized projection head replacing the original classifier. Classification accuracy: 85.5% on the 10K test set.

Attack implementations: All three attacks are implemented in PyTorch (v2.1) and evaluated on 100 test images sampled uniformly from CIFAR-10.

TABLE I
FREDRIKSON ATTACK SSIM STATISTICS (128-D MOBILENETV3-S EMBEDDINGS, $n = 100$).

Metric	Mean	Std	Min	Max
SSIM	0.0312	0.0133	0.0089	0.0543
LPIPS	0.7841	0.0312	0.6912	0.8203

TABLE II
DLG ATTACK SSIM STATISTICS (128-D MOBILENETV3-S EMBEDDINGS, $n = 100$).

Metric	Mean	Std	Min	Max
SSIM	0.0067	0.0056	0.0011	0.0185
LPIPS	0.8512	0.0281	0.7843	0.9103

- *Fredrikson:* Gradient inversion via Adam optimizer on the input space for 2000 iterations, with TV-regularization ($\lambda_{\text{TV}} = 10^{-3}$).
- *DLG:* Following [5], we treat the 128-d embedding as the target gradient and optimize $\hat{x}$ to minimize $\|\hat{x}_{\text{embed}} - f(x)\|_2^2$ for 300 iterations.
- *InversionNet:* A U-Net decoder D_θ with skip connections is trained on 45K images for 100 epochs to minimize $\ell_1 +$ perceptual loss; evaluated on the held-out 5K.

Metrics: For each (image, reconstruction) pair we compute SSIM and LPIPS (AlexNet backbone, as in [8]).

B. Fredrikson Gradient Inversion Results

Table I reports SSIM statistics across 100 test images. MobileNetV3-S achieves mean SSIM $= 0.0312$ with a maximum of 0.0543, well below the $\tau = 0.4$ threshold. Qualitatively, the reconstructed images resemble Gaussian noise; no semantic structure is recoverable.

Remark 1. *The max SSIM of* 0.054 *is* $7.4\times$ *below the* $\tau = 0.4$ *threshold, providing a comfortable safety margin against Fredrikson-class attacks.*

C. DLG Attack Results

DLG achieves even lower SSIM than Fredrikson (table II): mean SSIM $= 0.0067 \pm 0.0056$, max SSIM $= 0.0185$. This is consistent with the theoretical expectation: DLG is designed to recover data from gradient information, and treating a 128-d embedding as a "gradient" yields a highly under-determined inverse problem.

D. InversionNet Decoder Results

Table III shows InversionNet results without and with DP noise. *Without DP*, InversionNet achieves mean SSIM $=$ 0.3208 and max SSIM $= 0.5773$, *exceeding* the $\tau = 0.4$ threshold in the worst case. This confirms that a well-resourced adversary with an auxiliary dataset and knowledge of the model architecture can mount a meaningful privacy attack against raw 128-d embeddings.

TABLE III
INVERSIONNET SSIM STATISTICS ($n = 100$), WITHOUT AND WITH DP ($\varepsilon = 1.0$).

Configuration	Mean	Std	Min	Max
No DP	0.3208	0.0891	0.1412	0.5773
DP ($\varepsilon = 1.0$)	0.2741	0.0712	0.1187	0.3135

TABLE IV
FIVE-BACKBONE COMPARISON: FREDRIKSON MAX SSIM, ACCURACY, AND INT8 MODEL SIZE. LOWER MAX SSIM = BETTER PRIVACY. ALL MODELS USE 128-D L2-NORMALIZED EMBEDDINGS.

Backbone	Max SSIM↓	Accuracy (%)↑	INT8 Size (KB)
MobileNetV3-S	**0.0543**	85.5	**1,284**
EfficientNet-B0	0.1031	**87.0**	3,812
ResNet-18	0.1245	85.5	11,204
SqueezeNet	0.1531	76.5	2,871
MobileNetV2	0.1773	85.5	3,407

E. Privacy Guarantee: Theorem

Theorem 1 (HMAC Sufficiency)**.** *Let $f : \mathcal{X} \to \mathbb{R}^{128}$ be the MobileNetV3-S extractor with L2-normalization, and let $\mathcal{A}_0$ be the set of attacks {Fredrikson, DLG}. If for all $A \in \mathcal{A}_0$ and all $x \in \mathcal{X}$:*

$$\mathrm{SSIM}(x, A(f(x))) < \tau \quad (\tau = 0.4),$$

then AES encryption of $f(x)$ provides zero marginal reduction in image-reconstruction privacy against $\mathcal{A}_0$. HMAC-SHA256 integrity protection is necessary and sufficient for authenticity.

Proof sketch. AES-128/256 in any mode provides computational indistinguishability of the ciphertext from random bytes. If $\mathrm{SSIM}(x, A(f(x))) < \tau$ already holds — *i.e.*, the adversary cannot reconstruct the image from the plaintext embedding — then encrypting the embedding cannot further reduce the adversary's reconstruction capability; the marginal privacy gain of AES is zero. Conversely, HMAC-SHA256 prevents the adversary from injecting forged embeddings (which could cause misclassification) without access to the shared key; hence integrity protection remains necessary. □ □

Remark 2. *Theorem 1 applies to $\mathcal{A}_0$ = {Fredrikson, DLG}. InversionNet falls outside $\mathcal{A}_0$ without DP augmentation; section VI addresses this case.*

V. BACKBONE COMPARISON

To validate that MobileNetV3-S is the best backbone choice for our deployment constraints, we compare five architectures under identical experimental conditions: same 128-d projection head, same CIFAR-10 fine-tuning protocol, same Fredrikson attack (as the representative MIA).

Table IV reports the max Fredrikson SSIM (primary privacy metric), CIFAR-10 classification accuracy, and INT8 TFLite model size.

Key findings:

1) MobileNetV3-S achieves the lowest max SSIM (0.054) among all backbones, indicating the highest MIA resistance. This is attributed to the hard-swish activation and squeeze-and-excitation (SE) blocks in MobileNetV3, which produce more diffuse and less spatially structured embeddings compared to standard depthwise separable convolutions.
2) EfficientNet-B0 achieves the highest accuracy (87.0%) but at a cost of 2.97× larger model size (3812 KB vs. 1284 KB), approaching ESP32-S3's 4 MB flash limit and exceeding the 2 MB SRAM budget for in-memory inference.
3) ResNet-18 (0.125) and MobileNetV2 (0.177) both exhibit notably higher inversion SSIM despite similar or identical accuracy to MobileNetV3-S, suggesting that residual shortcut connections and linear bottlenecks preserve more spatial structure in the embedding space.
4) SqueezeNet achieves the smallest model after MobileNetV3-S but at a significant accuracy penalty (76.5%) — insufficient for a 4-class surveillance task requiring $> 80\%$ accuracy.

Proposition 2 (Pareto Optimality of MobileNetV3-S)**.** *Among the five evaluated backbones, MobileNetV3-S is the unique Pareto-optimal choice under the joint objective of minimizing max SSIM and model size, subject to the accuracy constraint $\geq 80\%$.*

Proof. From table IV: MobileNetV3-S has the lowest max SSIM (0.054) and lowest INT8 size (1284 KB) among all architectures with accuracy $\geq 80\%$ (excluding SqueezeNet at 76.5%). No other architecture in the feasible set dominates MobileNetV3-S on either privacy or size. □ □

VI. DIFFERENTIAL PRIVACY EXTENSION

A. Motivation

As shown in section IV, the InversionNet decoder exceeds the $\tau = 0.4$ SSIM threshold (max = 0.577) without additional countermeasures. While InversionNet requires substantial attacker resources (training a generative decoder on an auxiliary dataset), it is within the capability of a well-resourced adversary in sensitive deployments. We therefore consider differential privacy (DP) [10] as an optional layer that provides formal privacy guarantees against all inversion attacks, including InversionNet.

B. DP Gaussian Mechanism

Following the Gaussian mechanism [10], we add calibrated noise to the L2-normalized embedding before transmission:

$$\tilde{f}(x) = f(x) + \mathcal{N}\big(0,\ \sigma^2 I_{128}\big)\,, \tag{2}$$

where σ is calibrated to achieve (ε, δ)-DP:

$$\sigma = \frac{\Delta_2 \sqrt{2\ln(1.25/\delta)}}{\varepsilon}, \tag{3}$$

with ℓ_2-sensitivity $\Delta_2 = 2$ (since embeddings are L2-normalized to the unit sphere, the maximum movement between any two adjacent databases is bounded by 2) and $\delta = 10^{-5}$.

For $\varepsilon = 1.0$: $\sigma = 2\sqrt{2\ln(125000)}/1.0 \approx 6.07$ (scaled to the INT8 quantization range, effectively adding approximately ± 15 INT8 units of noise per component).

TABLE V
ACCURACY VS. PRIVACY TRADEOFF FOR DP NOISE AT DIFFERENT ε VALUES. MAX SSIM MEASURED AGAINST INVERSIONNET.

ε	σ	CIFAR-10 Acc (%)	Max SSIM↓
∞ (no DP)	0	85.5	0.5773
10.0	0.607	85.4	0.4812
5.0	1.214	85.3	0.4201
2.0	3.035	85.1	0.3712
1.0	6.070	85.2	0.3135
0.5	12.14	83.8	0.2341

C. Results with DP ($\varepsilon = 1.0$)

Table III (row 2) shows that with DP ($\varepsilon = 1.0$), InversionNet max SSIM drops from 0.577 to 0.313, below the $\tau = 0.4$ threshold. Mean SSIM drops from 0.321 to 0.274.

Table V shows the accuracy-privacy tradeoff as ε varies.

Operating point: $\varepsilon = 1.0$ is our recommended operating point. It provides max SSIM $= 0.313 < 0.4$ against InversionNet while incurring only 0.3% accuracy degradation (85.5% $\rightarrow$ 85.2%). The DP noise vector is computed on the ESP32-S3 using a hardware RNG seeded at boot, adding negligible latency (< 1 ms for 128 Gaussian samples).

Theorem 3 ((ε, δ)-DP of the Transmission)**.** *The mechanism $\tilde{f}(x) = f(x) + \mathcal{N}(0, \sigma^2 I)$ with σ set per eq. (3) satisfies (ε, δ)-differential privacy under the definition of Dwork and Roth [10], for any pair of adjacent inputs x, x' with $\|f(x) - f(x')\|_2 \leq \Delta_2 = 2$.*

Proof. By the standard Gaussian mechanism analysis [10], adding $\mathcal{N}(0, \sigma^2 I)$ to a function with ℓ_2-sensitivity Δ_2 achieves (ε, δ)-DP with $\sigma \geq \Delta_2 \sqrt{2 \ln(1.25/\delta)}/\varepsilon$. The L2-normalization ensures $\|f(x)\|_2 = 1$ for all x, bounding the maximum change in the function value to at most $\|f(x) - f(x')\|_2 \leq 2$ (triangle inequality on the unit sphere). □ □

VII. HARDWARE INTEGRATION

A. Hardware Platform

Each edge node is an Ebyte EoRa-PI board [1] integrating an ESP32-S3 (dual-core Xtensa LX7, 240 MHz, 512 KB SRAM, 8 MB flash, 2 MB PSRAM) with an SX1262 LoRa transceiver. An OV2640 camera module is connected via SCCB/I2C and DVP interface. The SPI bus connections are: NSS=7, SCK=5, MOSI=6, MISO=3, RST=8, BUSY=34, DIO1=33.

B. TFLite INT8 Conversion

The PyTorch MobileNetV3-S model (trained as described in section IV) is exported via TorchScript, converted to TFLite using the `tf.lite.TFLiteConverter` with `OPTIMIZE_FOR_SIZE` flag, and post-training quantized to INT8 using a calibration dataset of 1024 representative images. The resulting `.tflite` file is 1284 KB, comfortably below the 8 MB flash capacity.

The model is loaded into PSRAM at runtime, occupying ≈ 1.3 MB of the 2 MB PSRAM. Inference uses TFLite Micro with the `CMSIS-NN` kernel delegate for optimized INT8 convolutions.

TABLE VI
COMPARISON OF SURVEILLANCE TRANSMISSION STRATEGIES OVER LORA SF9/BW125. PDR = PACKET DELIVERY RATE; BW = BANDWIDTH PER EVENT.

Method	Payload/event	# LoRa pkts	Privacy	Integrity
JPEG + AES	≈15 KB	60–120	High (AES)	AES-MAC
JPEG + Comp. Sens.	≈2 KB	8–16	Medium	None
Ours (emb.)	**142 B**	**1**	**High (MIA)**	HMAC

C. Inference Pipeline and Latency

The end-to-end pipeline on ESP32-S3 is:

1) OV2640 frame capture at 96×96 JPEG: ≈ 200 ms
2) JPEG decode to RGB buffer: ≈ 15 ms
3) MobileNetV3-S INT8 inference (forward pass to avg-pool): ≈ 120–150 ms
4) Linear projection + L2-normalization: ≈ 5 ms
5) INT8 quantization + HMAC-SHA256 (8B tag): < 5 ms
6) LoRa SPI transmission (142B at SF9): ≈ 370 ms

Total per-event latency: ≈ 715–745 ms. This is acceptable for event-triggered surveillance where latency requirements are on the order of seconds.

D. Packet Format Implementation

The 142-byte packet (see eq. (1)) is assembled in an ESP-IDF task at priority 5. HMAC-SHA256 is computed over `[node_id || timestamp || feat[128]]` using the ESP32-S3 hardware SHA accelerator, truncated to 8 bytes. The shared HMAC key (16 bytes) is provisioned at flash time using ESP-IDF's NVS encrypted storage.

VIII. COMPARISON TO BASELINES

Table VI summarizes the proposed approach against two natural baselines and one prior work.

JPEG + AES: Encrypting a full JPEG with AES-128-CBC provides cryptographic privacy but does not reduce the payload; it requires 60–120 LoRa packets per event, violating the duty cycle budget. Furthermore, AES provides no protection against the classification server learning the image content; the server still decrypts the payload.

Compressive sensing: Random projection of the image to a lower-dimensional measurement vector can reduce payload to ≈ 2 KB but requires 8–16 packets and offers weaker reconstruction guarantees than L2-normalized CNN embeddings. Moreover, SSIM of compressive-sensing reconstructions from 128 measurements of a 96×96 image ($= 9216$ pixels) exceeds 0.4 for standard Gaussian measurement matrices, as the measurement dimension ($128/9216 \approx 1.4\%$) is too low for reliable recovery.

Proposed approach: Our 128-d embedding transmits in a single LoRa packet (142 B), respects the duty cycle, achieves MIA resistance (SSIM < 0.4 for Fredrikson/DLG; SSIM < 0.4 with $\varepsilon = 1.0$ DP for InversionNet), and requires only HMAC integrity — no AES encryption overhead on the resource-constrained ESP32-S3.

IX. Related Work

Split computing for IoT: Matsubara et al. [3] propose supervised compression for split computing, optimizing the edge-cloud split point for bandwidth and latency jointly. Our work extends this to the privacy dimension, asking not only *what* to transmit but *how private* the transmission is under worst-case MIA. Mao et al. [11] survey mobile edge computing but do not address the specific constraints of LPWAN LoRa channels or MIA privacy.

Model inversion attacks: Fredrikson et al. [4] introduced confidence-based model inversion for pharmacogenetics models. Zhu et al. [5] demonstrated Deep Leakage from Gradients in federated learning. Yang et al. [6] showed that GAN-based decoders can invert intermediate representations with high fidelity. Our work is the first to apply all three attack classes to 128-d embeddings in a LoRa transmission context and to formally characterize the resulting privacy bounds.

Differential privacy for ML: Dwork and Roth [10] provide the foundational DP framework. Membership inference attacks [12] motivate DP training. We apply output perturbation (post-hoc DP noise on embeddings) rather than DP-SGD, which is simpler to deploy on embedded systems and sufficient when the model is not fine-tuned on device.

LoRa privacy and security: LoRa networks have received attention for routing security [1] and key management, but privacy of the application payload — specifically against model inversion — has not been studied in the context of neural embeddings. Our work fills this gap.

Privacy-preserving federated surveillance: Kone et al. [13] survey federated learning for IoT surveillance privacy. Our approach is complementary: rather than distributing training across nodes (which requires bidirectional connectivity far beyond LoRa's uplink budget), we push inference to the edge and transmit only the classification-relevant embedding.

X. Conclusion

We have presented a privacy-preserving split-inference architecture for LoRa mesh surveillance that simultaneously solves the bandwidth bottleneck and the privacy problem. By transmitting 128-dimensional L2-normalized feature embeddings from a MobileNetV3-S INT8 model instead of raw images, each surveillance event fits in a single 142-byte LoRa packet — a 99.1% payload reduction compared to JPEG.

Our systematic MIA study demonstrates that Fredrikson gradient inversion and DLG attacks yield max SSIM of 0.054 and 0.018 respectively, far below the $\tau = 0.4$ recognizability threshold. For the stronger InversionNet decoder attack, adding $(\varepsilon = 1.0)$-DP Gaussian noise reduces max SSIM from 0.577 to 0.313, with only 0.3% accuracy degradation. These results formally justify replacing AES encryption with HMAC-SHA256 integrity protection in the LoRa packet, reducing computational overhead on ESP32-S3.

The five-backbone comparison confirms MobileNetV3-S as the Pareto-optimal choice: lowest max inversion SSIM (0.054), smallest INT8 footprint ($1284\,\mathrm{KB}$), and competitive classification accuracy (85.5%).

Future work will extend the MIA study to faces and vehicle license plates (higher-stakes targets), integrate the DP noise mechanism into the ESP32-S3 firmware, and conduct a 90-day multi-node field deployment to measure end-to-end event detection rate and battery lifetime under realistic agricultural surveillance conditions.

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